TEZPUR UNIVERSITY

Spring Semester End Examination, 2024

Course Code: CO206 **Course Title:** Design and Analysis of Algorithms

Full Marks: 60 **Time:** 3 hours

You may answer all the questions but maximum you can get is 60.

1. Following is a version of vertex cover problem.

Input: Undirected graph $G = (V, E)$, and an integer k.
Output: (yes/no:) Does G have a vertex cover of size k?

A vertex cover is a set $U \subseteq V$ such that every edge in E has at least one endpoint from the set U. Does the above problem belong to the class NP? [5]

2. Suppose you have to fill up an array of size n with 5 numbers: 1,2,3,4,5.

   a) You have to fill the array in a way such that consecutive elements are different. Write a recurrence relation to count the number of such arrangements. Write the base cases. [5]

   b) Say you decide not to have have same 3 numbers consecutively. Write a new recurrence relation and provide base case(s) for this new restriction. [7]

3. Answer True / False and provide a counterexample or a short justification. [8×3]

   a) The shortest path between two vertices in a weighted graph is unique if the weights of edges are distinct.

   b) $n^{100} + 1.01^n \in O(n^{101})$.

   c) Dijkstra's shortest path algorithm works perfectly for any graph which contains exactly one negative edge.

   d) Any graph with n vertices and atleast $\frac{n(n-1)}{4}$ edges is a connected graph.

   e) The 2nd largest weighted edge of a graph G will be in the minimum spanning tree of G.

   f) The SORTING problem belongs to the class NP.

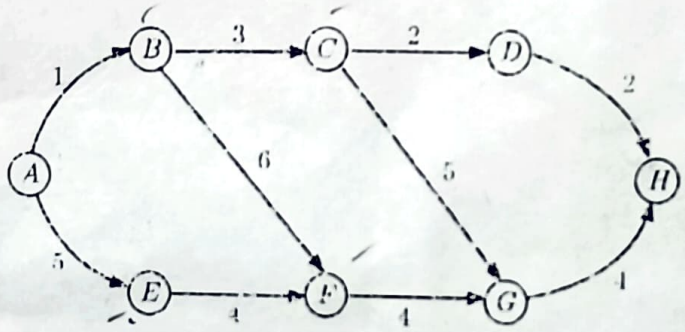
   g) If $f(n) = O(g(n))$ and $g(n) = O(h(n))$, then $h(n) = \Omega(f(n))$.

   h) Suppose $f(i) = f(i-1) + 2f(i-2) + 3f(i-3)$ and $f(0) = 1, f(1) = 2, f(2) = 3$. We can determine the value of $f(n)$ in $O(n)$ time.

4. Consider an instance of 0/1 Knapsack problem where all the items have same values but have distinct weights. If capacity of the Knapsack is C, can we design a greedy algorithm to choose the most valuable subset of items whose total weight does not exceed C? If yes, write the algorithm and argue it's correctness. [5+5]

5. Suppose we have a bag of capacity 10 kg, which we want to pack with items of weights 2 kg, 3 kg, 4 kg and 5 kg of values 3, 4, 5 and 7 respectively. Our goal is to pack the bag with as many as items possible so that the total weight of the items does not exceed the capacity of the bag and the profit is maximized. Solve the problem using dynamic programming **(Show the table only)**. Explain the running time of the algorithm. [7]

6. Suppose we run Dijkstra's single source shortest-path algorithm on the following edge weighted directed graph with vertices A as the source. For the graph below, find the order in which Dijkstra's algorithm will compute the distances of the vertices, starting from vertex A. Show the status of the HEAP in each iteration, showing the "score" of each vertex. What will be distances of all the vertices from the source vertex A? [10]



7. Draw the Huffman tree corresponding to the encoding table below. What will be the range of the possible values for the frequency of the character 'P'? Explain why. [2+3]

symbol	C	O	M	P	U	T	E
frequency	2	1	3	?	5	15	15
code	01111	01110	0110	00	010	10	11